



Patricia Rodrigo Barrio

Data Analyst | Machine Learning | Process Improvement

Biomedical Engineer and MSc student in Data Analysis Engineering, Process Improvement and Decision Making at Universitat Politècnica de Valencia. My profile combines data analysis, machine learning, deep learning and computational modelling. I am especially interested in using artificial intelligence to transform complex data into useful insights for decision-making, both in clinical and biomedical environments and in industrial contexts.

I consider myself an analytical, curious and persistent person, with a strong ability to learn new tools, document technical work rigorously and turn complex technical problems into clear, reproducible and results-oriented solutions.

CONTACT

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- LinkedIn: Patricia Rodrigo Barrio
- GitHub: prodbar
- Portfolio: prodbar

EDUCATION

- Universitat Politècnica de Valencia
MSc in Data Analysis Engineering, Process Improvement and Decision Making
Sep. 2025 - Present
- Universitat Politècnica de Valencia
BSc in Biomedical Engineering
Sep. 2021 - Jun. 2025
GPA: 8.2/10 - Final Degree Project with Honours

PROJECTS AND PORTFOLIO

- Portfolio with projects in data analysis, machine learning, visualisation, optimisation and computational modelling.
- GitHub repositories with code, notebooks, technical documentation and reproducible results.

LANGUAGES

Spanish - Native
English - C1 (Cambridge CAE)

AWARDS

May 2021 | First prize

VI Virtual Agri-food, Agri-environmental and Forestry Olympiad of Castilla y Leon

EXPERIENCE

I2MB - Institute of Mechanical and Biomechanical Engineering

Oct. 2024 - Present

Valencia, Valencian Community (Spain)

Clinical Data Analyst - Research Trainee

- Analysis, cleaning and structuring of real clinical data in biomedical engineering projects.
- Development of metrics, indicators and visualisations to support result interpretation and decision-making.
- Participation in PREVILI, focused on hospital clinical data analysis in neonatal and paediatric patients.
- Development of PINNs for hyperelastic biomechanical modelling, integrating deep learning, physical constraints and FEM comparison.

Morcillas Riostras Gourmet

Jul. 2024 - Sep. 2024

Burgos, Castile and Leon (Spain)

Production Processes and Quality Control Support

- Awarded the Campus Rural 2024 scholarship.
- Support in production, operational organisation and quality control in an industrial environment.
- Applied an analytical approach to real problems in organisation, quality and process management.

TECHNICAL SKILLS

Programming and tools: Python · SQL · MATLAB · R · Advanced Excel · Power BI · OpenSolver.

Data analysis: data cleaning · variable transformation · metrics generation · results evaluation.

Machine Learning and Deep Learning: predictive models · classification · model validation · neural networks · hyperparameter analysis.

Biomedical processing: clinical data · signal processing · image processing · feature extraction.

Processes and decision-making: quality control · operational analysis · process improvement · decision support.

Computational modelling: FEM · PINNs · numerical simulation · hyperelasticity · forward and inverse problems.